

Galois 1

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5. Fixed Field

Claim 0. $\forall p(t) \in C(t)$, where $p(t) \notin \mathbb{C}$, $\mathbb{C}(p(t)) = \mathbb{C}(t)$

Proof. $\mathbb{C}(p(t)) \subseteq \mathbb{C}(t)$ because $p(t) \in \mathbb{C}(t)$. □

Problem 5.1a. $\sigma(t) = t^{-1}$

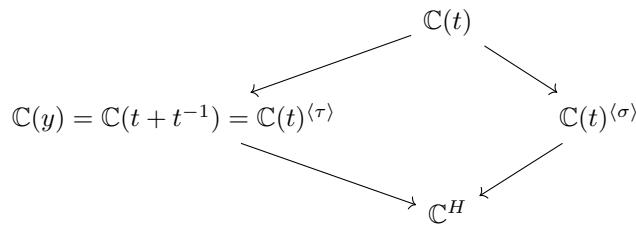
Proof. $H = \langle \sigma \rangle = \mu_2$. $p_t(x) = \prod_{\alpha \in H \cdot t} (x - \alpha) = (x - t)(x - t^{-1}) = x^2 - x(t + t^{-1}) + 1$ is the minimal polynomial of t over $\mathbb{C}(t)^H$. By the previous homework, $\mathbb{C}(t)^H = \mathbb{C}(p_t) = \mathbb{C}(t + t^{-1})$ □

Problem 5.1b. $\sigma(t) = it$

Proof. The orbit of t under σ is $t, it, -t, -it, t, \dots$. Thus, $\langle \sigma \rangle$ has order 4. Thus $H = \langle \sigma \rangle = \mu_4$. $p_t(x) = (x - t)(x + t)(x - it)(x + it) = (x^2 - t^2)(x^2 + t^2) = x^4 - t^4$ over $\mathbb{C}(t)^H$. By the previous homework, $\mathbb{C}(t)^H = \mathbb{C}(p_t) = \mathbb{C}(t^4)$ □

Problem 5.1c. $\sigma(t) = -t$ and $\tau(t) = t^{-1}$.

Proof. Orbit of t under σ is $t, -t, t, \dots$. The orbit of t under t^{-1} is $t, t^{-1}, t, \dots$. Thus, $o(\sigma) = 2$ and $o(\tau) = 2$. Further more, $(\sigma \circ \tau)(t) = \sigma(t^{-1}) = -t^{-1} = \tau(-t) = (\tau \circ \sigma)(t)$. Thus $\sigma\tau = \tau\sigma$. Thus $H = \langle \sigma, \tau \rangle = \mu_2 \times \mu_2$



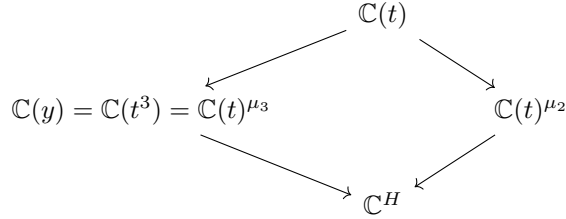
Since $\langle \sigma, \tau \rangle$ is abelian, $\langle \tau \rangle \triangleleft \langle \sigma, \tau \rangle$, $\langle \sigma \rangle$ preserves $\mathbb{C}(y)$ thus $\langle \sigma \rangle \subset \text{Gal}(\mathbb{C}(y)/\mathbb{C})$. $\mathbb{C}^H = \mathbb{C}(y)^{\langle \sigma \rangle}$. $p_y(x) = (x - y)(x + y) = x^2 - y^2$, min poly of y over $\mathbb{C}^H$. Thus $\mathbb{C}^H = \mathbb{C}(y^2) = \mathbb{C}((t + t^{-1})^2)$ □

Problem 5.1d. $\sigma(t) = \omega t$, $\gamma(t) = t^{-1}$ where $\omega = e^{2\pi i/3}$

Proof. Orbit of t under σ is $t, \omega t, \omega^2 t, t, \dots$. Thus, $o(\sigma) = 3$ and $o(\gamma) = 2$. $(\gamma \circ \sigma \circ \gamma^{-1})(t) = \gamma(\sigma(t^{-1})) = \gamma(\sigma(t)^{-1}) = \gamma(\bar{\omega} t^{-1}) = \bar{\omega} t = \sigma^{-1}(t)$. Thus $\gamma \circ \sigma \circ \gamma = \sigma^{-1}$. Thus $H = \langle \sigma, \gamma \rangle = D_3$.

Note: $\mu_3 = \langle \sigma \rangle$ and $\mu_2 = \langle \gamma \rangle$.

$p_t(x) = (x - t)(x - \omega t)(x - \omega^2 t) = x^3 + t^3$, min poly of t over $\mathbb{C}(t)^{\mu_3}$. Thus, $\mathbb{C}(t)^{\mu_3} = \mathbb{C}(t^3)$. Let $y = t^3$.



Since, $\mu_3 \triangleleft \langle \sigma, \gamma \rangle$, $\mu_2 \leq \text{Gal}(\mathbb{C}(t)^{\mu_3}/\mathbb{C})$. Thus, $\mathbb{C}^H = \mathbb{C}(y)^{\mu_2}$.

Orbit of y over μ_2 , is y and y^{-1} . Min poly of y over $\mathbb{C}^H$ is $p_y(x) = (x - y)(x - y^{-1}) = x^2 - x(y + y^{-1}) + 1$. Thus $\mathbb{C}(t)^H = \mathbb{C}(y + y^{-1}) = \mathbb{C}(t^3 + t^{-3})$ $\square$

Problem 5.2. Show that the automorphisms, $\sigma(t) = \frac{t+i}{t-i}$ and $\rho(t) = \frac{it-i}{t+1}$ of $\mathbb{C}(t)$ generate the group isomorphic to A_4 and determine the fixed field of this group

Proof. By the given hint, $\text{Gal}(\mathbb{C}(t)/\mathbb{C}) \cong PGL_2(\mathbb{C})$ and $PGL_2(\mathbb{C})$ acts simply transitively on any ordered triple. Thus there is a unique element of $\text{Gal}(\mathbb{C}(t)/\mathbb{C})$ which sends $a \rightarrow 1, b \rightarrow 0, c \rightarrow \infty$ and vice versa, (ie the unique ordered triple (a, b, c) where $1 \rightarrow a, 0 \rightarrow b$ and $\infty \rightarrow c$ determines a element of $PGL_2(\mathbb{C})$).

Thus check the action of σ and τ on 0 1 and ∞ .

Orbit of 0 over σ .

$$0 \xrightarrow{\sigma} -1 \xrightarrow{\sigma} -i \xrightarrow{\sigma} 0$$

Orbit of 1 over σ .

$$1 \xrightarrow{\sigma} i \xrightarrow{\sigma} \infty \xrightarrow{\sigma} 1$$

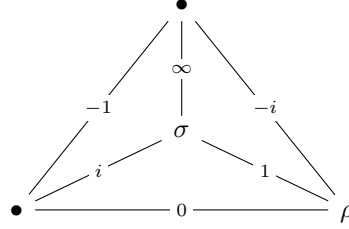
Orbit of 0 over ρ .

$$0 \xrightarrow{\rho} -i \xrightarrow{\rho} 1 \xrightarrow{\rho} 0$$

Orbit of -1 over ρ .

$$-1 \xrightarrow{\rho} \infty \xrightarrow{\rho} i \xrightarrow{\rho} -1$$

We have 6 elements which we are permuting, $0, 1, -1, i, -i, \infty$. We want to prove that the group of $\langle \sigma \rangle$ and $\langle \rho \rangle$ is A_4 so we want to equate these elements to something of the tetrahedron. We have 6 elements of a tetrahedron which we permute with A_4 , its edges. So place the elements on the edges of a tetrahedron as below. σ and ρ are placed on the vertices which they would be rotations through if they were rotations of this tetrahedron.



Thus $\langle \sigma, \rho \rangle \subseteq A_4$ and $9 \leq |\langle \sigma, \rho \rangle| \leq 12$ (two 3 cycles which are not in the subgroups generated by each other's elements). Thus $|\langle \sigma, \rho \rangle| = 12$ since $|\langle \sigma, \rho \rangle| \nmid 12$.

$\sigma(\rho(t)) = \sigma(\frac{it-i}{t+1}) = \sigma(it-i)/\sigma(t+1) = (i\sigma(t)-i)/(\sigma(t)+1) = \frac{\frac{it-1}{t-i}-i}{\frac{t+i}{t-i}+1} = -t^{-1}$. $\rho(\sigma(t)) = \rho(\frac{t+i}{t-i}) = \rho(t+i)/\rho(t-i) = (\rho(t)+i)/(\rho(t)-i) = \frac{\frac{t+1}{t-i}+i}{\frac{it-i}{t+1}-i} = -t$. $o(\sigma \circ \rho) = 2$ and $o(\rho \circ \sigma) = 2$. $((\sigma\rho) \circ (\rho\sigma))(t) = (\sigma\rho)(-t) = t^{-1}$ and $((\rho\sigma) \circ (\sigma\rho))(t) = (\rho\sigma)(-t^{-1}) = -((\rho\sigma)(t))^{-1} = t^{-1}$. Thus $(\rho\sigma) \circ (\sigma\rho) = (\sigma\rho) \circ (\rho\sigma)$. Thus $\langle \sigma, \rho \rangle = K_4$ and $\sigma\rho^2\sigma = t^{-1}$.

$\mathbb{C}(t)^{K_4}$ is preserved by all of A_4 since $K_4 \trianglelefteq A_4$. Thus $C(t)^{A_4} = (\mathbb{C}(t)^{K_4})^{\mu_3}$ where $\mu_3 = \langle \sigma \rangle = \langle \rho \rangle$.

Orbit of t over K_4 is $t, -t, -t^{-1}, t^{-1}$. Thus min poly of t over $\mathbb{C}(t)^{K_4}$ is $p_t(x) = (x-t)(x+t)(x+t^{-1})(x-t^{-1}) = (x^2-t^2)(x^2-t^{-2}) = x^4 - x^2(t^2-t^{-2}) + 1$. Thus $\mathbb{C}(t)^{K_4} = \mathbb{C}(t^2+t^{-2})$. Let $y = t^2+t^{-2}$.

$$\begin{aligned} \sigma(y) &= \sigma(t^2+t^{-2}) = \sigma(t)^2 + \sigma(t)^{-2} \\ &= \frac{2(1-6t^2+t^4)}{(1+t^2)^2} \\ &= 2(1 - \frac{8t^2}{t^4+2t^2+1}) \\ &= 2(1 - \frac{8}{t^2+t^{-2}+2}) \\ &= 2(1 - \frac{-8}{y+2}) \\ &= \frac{2(y+2)-16}{y+2} \\ &= \frac{2y-12}{y+2} \end{aligned}$$

You can convert σ 's action on y as a matrix, where the top row is the numerator and taking subsequent applications of the σ is the same thing as matrix multiplication. Moreover remember we are in $P(\mathbb{C})$ so if two matrices are scalar multiples of each other they are equal.

Thus $\sigma(y) = \begin{bmatrix} 2 & -12 \\ 1 & 2 \end{bmatrix}$ and checking it has order 3, we see $\sigma(t)^3 = \begin{bmatrix} -64 & 0 \\ 0 & -64 \end{bmatrix} \equiv I$.

$$\sigma^2(y) = \begin{bmatrix} -8 & -48 \\ 4 & -8 \end{bmatrix} \equiv \begin{bmatrix} -2 & -12 \\ 1 & -2 \end{bmatrix} = -2(\frac{y+6}{y-2})$$

Thus the orbit of y under σ is $y, \frac{2y-12}{y+2}, \frac{-2y-12}{y-2}$.

Thus $p_y(x) = (x-y)(x-\frac{2y-12}{y+2})(x-\frac{-2y-12}{y-2})$

$$\begin{aligned}
p_y(x) &= x^3 - \left(\frac{(-12-2y)y(-12+2y)}{(-2+y)(2+y)} \right) \\
&+ x \left(\frac{(-12-2y)y}{-2+y} + \frac{(-12-2y)(-12+2y)}{(-2+y)(2+y)} + \frac{y(-12+2y)}{2+y} \right) \\
&+ x^2 \left(-y - \frac{-12+2y}{2+y} + \frac{12+2y}{-2+y} \right)
\end{aligned}$$

Let C_{p_y} be the coefficients of p_y . Thus $\mathbb{C}(t)^{A_4} = \mathbb{C}(C_{p_y})$. □

Problem 5.3. Let $F = \mathbb{C}(t)$ be field of rational functions in t . Prove every element in F that is not in $\mathbb{C}$ is transcendental in $\mathbb{C}$.

Proof. Let $f(t) \in F \setminus \mathbb{C}$. $f(t) = \frac{p(t)}{q(t)}$ for $p(t), q(t) \in \mathbb{C}[t]$ where $\gcd(p(t), q(t)) = 1$. Let $n = \max\{\deg(p), \deg(q)\}$.

Let $g(X) \in \mathbb{C}(f(t))[X]$ be defined as follows,

$$g(X) = f(X)q(X) - p(X)$$

t is a solution to this polynomial. If all the coefficients of $g(x)$ are zero, take a nonzero coefficient a of q and the coefficients b of p of the same power. Then since all the coefficients are 0, then $f(X)a + b = 0 \implies f(X) = -\frac{b}{a} \in \mathbb{C}$ which contradicts our initial statement. Thus at least coefficients of $g(x)$ are nonzero. Thus t is algebraic over $\mathbb{C}(f(t)) \implies \mathbb{C}(t)$ is f.d over $\mathbb{C}(f(t))$.

Assume the contrary, that $f(t)$ is algebraic, thus $\mathbb{C}(f(t))$ is f.d over $\mathbb{C}$. By a previous homework (I think Fields 1), that means $\mathbb{C}(t)$ is fd over $\mathbb{C}$ which means that t is algebraic over $\mathbb{C}$ which is a contradiction. Thus $f(t)$ must be transcendental. □

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Problem 6. Let L be a primitive extension of K , $L = K(t)$ (not necessarily algebraic, we can allow $L =$ rational functions). Let G be a finite subgroup of $\text{Gal}(L/K)$, let $G \cdot t$ be the orbit, and $P(X)$ the poly in $L[X]$ with roots the elements of Gt . Let $K \subset W \subset L$ be the subfield generated by the coefficients of P . Show L over W is a primitive algebraic extension, generated by t , with minimal poly P . Show that W is the fixed field of G , and that L over W is galois with group G .

Proof. We have previously proven all of these statements. By the Finite Fields and Galois Correspondence homework, P is irreducible and separable and splits over L and thus the minimal polynomial of t over $W = K(P)$. Furthermore by the same homework, L is galois over $K(P)$. Furthermore by the Galois Correspondence homework, $L = W[t]$. Again by Finite Fields and Galois Correspondence, $\text{Gal}(L/W) = G$ and $W = L^G$. □